

Nikkat Afrin

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SUMMARY

AI & Data Platform Engineer with 2+ years of experience building production speech-AI, RAG, cloud-native applications, and analytics pipelines. Hands-on expertise in Python, FastAPI, LangChain, PyTorch, SQL, AWS, Azure, Databricks, and vector search; first-author IEEE researcher focused on scalable, measurable AI and data systems.

PROFESSIONAL EXPERIENCE

AI Platform Engineer, Peblink - World Literacy Research Center | New York, NY Jan 2026 - Present

- Achieved 93% pronunciation-detection accuracy across 12K+ student reading sessions by engineering Whisper-based speech assessment pipelines using FastAPI and transcript-alignment algorithms.
- Reduced literacy-assessment review turnaround from 60 minutes to under 5 minutes by deploying AWS serverless pipelines processing multilingual audio-video datasets through S3, Lambda, and DynamoDB.
- Enabled research analytics by transforming speech-assessment outputs into S3 Parquet and DynamoDB datasets queryable through Amazon Athena and visualized in QuickSight.

AI/Data Analyst Intern, Peblink - World Literacy Research Center | New York, NY May 2025 - Jan 2026

- Architected the WLRC global literacy AI/data hub within eight weeks by designing AWS ingestion pipelines, role-based workflows, and analytics dashboards supporting universities and NGOs.
- Built literacy benchmarking frameworks across 30+ public datasets, identifying programs delivering 15–20-point reading improvements using Python, Tableau, and comparative educational analytics methodologies.
- Collaborated with Peblink leadership to translate literacy analytics into AI product requirements, supporting partnership strategy and discussions with Congressman Brad Schneider and nonprofit stakeholders.

AI Workflow Engineer Intern, KAYAAN Inc. (Logistics AI Startup) | New York, NY Aug 2025 - Dec 2025

- Increased investor outreach effectiveness by automating AI-agent workflows, researching 500+ supply-chain investors, achieving 24% connection rates, and 8% follow-up engagement through personalized messaging automation.
- Built React dashboards helping leadership monitor 15+ logistics, operational, and profitability KPIs across shipment activity, platform engagement, and product-performance workflows.
- Improved deployment readiness by validating 20–40 QA tickets per sprint, reproducing workflow defects, and accelerating engineering issue-resolution turnaround before production releases.

Graduate Assistant - AI & Data (Part-time), Yeshiva University, Shevet Glaubach Career Center | New York, NY Jan 2025 - Dec 2025

- Delivered real-time academic-support assistance for 2K+ students by co-developing FELIX RAG chatbot using LangChain, ChromaDB, OpenAI GPT-4/3.5, Flask, and Azure services.
- Eliminated 40+ weekly staff reporting hours and improved advising productivity 12% by automating Symplicity CSM ingestion through Azure Functions, Databricks, and PostgreSQL pipelines.

Graduate Assistant - AI & Admissions (Part-time), Yeshiva University, Katz School of Science & Health | New York, NY Jun 2024 - Aug 2025

- Accelerated applicant-response workflows by building AI-powered admissions assistants generating contextual inquiry responses, reusable email templates, and applicant-support knowledge automation capabilities.
- Improved admissions decision-making visibility by developing Power BI dashboards tracking applicant funnels, outreach conversion trends, enrollment pipelines, and Recruit Ellucian engagement metrics.
- Organized admissions-response knowledge resources and reusable templates, improving communication consistency across graduate-assistant support workflows.

PUBLICATION

SigmaCam: Exact Decision Boundary Extraction for DNNs (First Author), IJCNN 2025 (IEEE Xplore) Mar 2025

- Improved AI model interpretability by introducing SigmaCam, a PyTorch-accelerated algorithm that extracts exact decision boundaries for smooth neural-network activations like Sigmoid and SiLU, reducing runtime by 93% while maintaining 99.8% accuracy.

PROJECTS

Azure Real-Time Voice Conversational AI [GitHub](#)

- Built a low-latency speech-to-speech RAG platform using Azure OpenAI, Azure Speech, Azure AI Search, FastAPI, WebSockets, and Container Apps.

Amazon Connect Real-Time Chat Translation [GitHub](#)

- Developed bidirectional multilingual chat translation using Amazon Connect, Amplify, Lambda, Translate, Comprehend, API Gateway, and React.

Disease Management Data Platform [GitHub](#)

- Designed PostgreSQL OLTP and dimensional models with SCD Type 2, Neo4j graph relationships, and SQL analytics for healthcare reporting.

CERTIFICATIONS

Google Cloud Certified Professional Data Engineer [Link](#)

Aug 2026 - Aug 2028

Microsoft Certified: Fabric Data Engineer Associate [Link](#)

Aug 2026 - Aug 2027

EDUCATION

Yeshiva University, Katz School of Science & Health | New York, USA | Master of Science, Data Analytics & Visualization | GPA: 3.95 Jan 2024 - Jan 2026

Andhra University | Visakhapatnam, India | Bachelor of Computer Applications | GPA: 4.0/4.0

Jun 2019 - Jul 2022

SKILLS

AI/ML & Generative AI: LLMs, RAG, LangChain, ChromaDB, OpenAI APIs, GPT-4/4o, Prompt Engineering, Whisper, faster-Whisper, Hugging Face, PyTorch, TensorFlow, XGBoost, SHAP, Speech AI, Vector Search, AI Agents, NLP, Deep Learning

Cloud, Backend & MLOps: AWS (Lambda, S3, SQS, DynamoDB, Athena, QuickSight, API Gateway, Cognito, EC2, IAM), Azure OpenAI, Azure AI Search, Azure Speech, Azure Functions, Blob Storage, Container Apps, FastAPI, Flask, REST APIs, WebSockets, Docker, Terraform, GitHub Actions

Data Engineering & Analytics: Python, SQL, Pandas, NumPy, scikit-learn, Databricks, Apache Spark, PySpark, Delta Lake, ETL Pipelines, PostgreSQL, MySQL, MS SQL Server, Neo4j, Data Modeling, Power BI, Tableau, Excel

Frontend & Automation: React.js, JavaScript, TypeScript, R, n8n, Workflow Automation, QA Testing, Google Sheets, Git/GitHub, API Integrations

AWARDS

Best Research Project Award, Yeshiva University CSE Research Forum, 2025 - SigmaCam research.

Innovation Lab Challenge Finalist, Yeshiva University, 2025 - selected as 1 of 7 teams from 30.

Gold Medalist / University Topper, Andhra University - graduated with a 4.0/4.0 GPA.